Inspectable Clinical Extraction from Epilepsy Letters

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Abstract—**Clinical extraction results are difficult to interpret when model output, deterministic rules, repair, evidence checks, and scoring are reported as one method. We evaluate three methods on two epilepsy-letter tasks: current seizure-frequency labeling on the Gan 2026 synthetic benchmark and broad epilepsy phenotyping on ExECTv2. For each task we compare rules only, LLM only, and LLM with rules. Saved Gan holdout results are 364/450 Purist for the single-pass system and 379/450 for a multi-model comparison. On ExECT dev140, rules only reached 0.6020 paper-derived all-features macro item F1 (0.3548 under the existing strict micro scorer), the LLM only method reached 0.7393 clinical fact F1. The historical combined control reached 0.9189 but used deterministic Prescription and SF-union paths; final six-model results are pending. Saved-output ablations show positive normalization contributions on both tasks, while the evidence check is score-inert on those representative replays. The selected evidence supports a reproducible component comparison with explicit data limits and a tested implementation of the published ExECT metric views. It does not reproduce the original ExECT system or its reported 0.87/0.90 scores, and it does not support independent clinical validation or a completed six-model result.**

Index Terms—**clinical information extraction, epilepsy, large language models, reproducibility, evidence attribution**

I. Introduction

Epilepsy letters combine temporal reasoning, clinical terminology, medication regimens, investigation findings, and ambiguous current-versus-historical statements. Large language models (LLMs) can broaden extraction, but a model call can hide where a result came from. Deterministic systems are easier to inspect, yet often fail on linguistic variation and long-range clinical selection.

We compare rules-only, LLM-only, and LLM-with-rules methods. The same package contains a deep single-label task and a broad multi-entity task. This makes it possible to ask which stages transfer, which results depend on task-specific scoring, and where evidence is insufficient for a stronger claim.

The paper contributes: (1) selected results for three methods on two tasks; (2) explicit ownership of extraction, normalization, final formatting, repair, evidence checking, and scoring; (3) saved aggregate Gan holdout results and partial ExECT model-transfer evidence; and (4) reproducibility controls that pin prompts, scorers, splits, repair policies, runtime identifiers, dependencies, and file hashes.

II. Related Work

ExECT demonstrated structured epilepsy extraction from unstructured clinic letters and later work documented the corresponding annotation resource [1], [2]. Seizure-frequency extraction has also been studied with machine-reading and pretrained-language-model approaches [3], [4], while recent work has explored fine-tuned LLMs and reproducible synthetic clinical letters [5], [6]. Our focus is complementary: processing steps and data limits are part of the scientific method, and rules-only, LLM-only, and LLM-with-rules methods are compared without collapsing their different scoring and repair mechanisms.

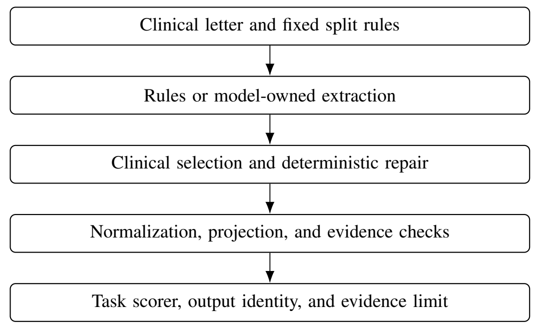

Fig. 1. Processing steps and their owners. Ownership remains explicit when a step is implemented by an LLM.

III. Methods

A. Tasks and Split Policy

Gan 2026 asks for one current seizure-frequency label per synthetic letter [6]. Validation750 is the row-inspectable development and replay split. Test450 is a locked holdout: only saved aggregate results are retained, and row-level holdout failures are not used for development.

ExECTv2 covers Diagnosis, SeizureFrequency, Prescription, and Investigations in the primary comparison, with additional entities retained in the deterministic all-nine reference [1], [2]. Dev140 is the row-inspectable development split. Full200 combines dev140 with held-out test60 and is reported only as a development-inclusive aggregate audit; it is not an independent holdout.

B. Compared Methods

For each task, the retained evidence index selects one rules-only, LLM-only, and LLM-with-rules run. Rules-only runs use no model calls. LLM-only runs put the clinical decision in one model program. The ExECT GEPA run is a negative development comparison. The final ExECT comparison requires model-origin facts in all four families and prohibits independent Prescription or Seizure Frequency substitution or union.

TABLE I
SELECTED RESULTS FOR THREE METHODS ON TWO TASKS

Task	Method	Split	Selected result	Use and limit
ExECTv2	Rules only	dev140	All-feature macro item F1 0.6020	Paper-derived development result; strict micro F1 0.3548.
ExECTv2	LLM only	dev140	Headline F1 0.7393	GEPA negative development comparator.
ExECTv2	Historical LLM with rules	dev140	Clinical fact F1 0.9189	Reproducible control; not the final model-led architecture.
Gan 2026	Rules only	validation750	697/750 Purist	Deterministic development comparator.
Gan 2026	LLM only	validation750	581/750 Purist	Single-pass development comparator.
Gan 2026	LLM with rules	validation750	661/748 rendered Purist	Single-pass event comparison.

TABLE II
EXECT RULES-ONLY DEV140 METRIC VIEWS

View	Per-item F1	Per-letter F1
Normalized phrase	0.5687	0.7518
CUI	0.7144	0.8534
All features	0.6020	0.7922

TABLE III
GAN SAVED RESULTS ON LOCKED TEST450

System	Purist	Role
Single structured-event pass	364/450	One model pass per note
V12 multi-trace reviewer	379/450	Three model passes per note

TABLE IV
PLANNED EXECT SIX-MODEL COMPARISON

Model condition	Overall	Diagnosis	SF	Prescription	Investigations	Call/parse failures
GPT-4.1-mini	Pending	Pending	Pending	Pending	Pending	Pending
GPT-5.6 Luna	Pending	Pending	Pending	Pending	Pending	Pending
GPT-5.6 Sol	Pending	Pending	Pending	Pending	Pending	Pending
DeepSeek V4 Flash (thinking)	Pending	Pending	Pending	Pending	Pending	Pending
Qwen 3.6:35B (local)	Pending	Pending	Pending	Pending	Pending	Pending
Gemma 4 26B (local)	Pending	Pending	Pending	Pending	Pending	Pending

C. Scoring, Repair, and Attribution

Gan uses Purist and Pragmatic label scoring; Purist is the primary strict label score. ExECT's primary internal score is de-duplicated `clinical_headline` recovery across four families. Phrase, CUI, evidence-valid, and full-attribute companions are reported separately. `clinical_headline` is not presented as reproduction of the published strict ExECT benchmark. The paper-derived views score each entity type separately and report their macro mean. Per-item scoring counts mentions; per-letter scoring requires at least one correct mention and feature bundle. Certainty applies to Diagnosis and PatientHistory; negation applies only to PatientHistory.

Raw model selection, JSON or format repair, source-exact evidence repair, deterministic semantic repair, projection, and final scoring are separate stages. Model-specific transport or output-shape adapters do not become model quality evidence. Semantic repairs retain a named deterministic owner and must be ablated or otherwise tested before supporting a causal interpretation.

D. Recorded Reference Version

The retained evidence index records implementation commit `46562134`, the dependency declaration and lock, prompts, scorers, split definitions, repair policies, model policy, split runbook, and CI workflow. A semantic prompt, scorer, split, repair, runtime-route, or component-graph change requires a new recorded version and complete replay. This rule does not authorize model calls.

IV. RESULTS

A. Selected Method Comparison

Table I is the minimum selected scientific comparison. Scores are not compared across tasks because the datasets, outputs, and scorers differ.

All six runs replay from current code and selected outputs without model calls. The ExECT combined-method replay also returns evidence-valid F1 0.8913. Its legacy benchmark/CUI companion is 0.4791 versus 0.4729 in the saved run; this diagnostic does not define the paper-derived rules-only result or affect the reproduced 0.9189 headline score. Its deterministic Prescription producer and SF extractor union are architecture limits, not model contributions.

B. ExECT Paper-Derived Metric Replay

The no-call replay covers all nine entity types. CUI identity recovers many surface-form misses, while attribute agreement—especially for Diagnosis—is the main remaining loss. One gold mention and six predictions lack a CUI; missing identifiers never match. The original 0.87 per-item and 0.90 per-letter scores remain reference values, not reproduced results.

C. Gan Locked Holdout Evidence

Both rows in Table III are saved aggregate evidence. The 15-row quality difference is 3.33 percentage points. A cold execution requires one model pass per note for the single-pass system and three for the saved V12 test condition. DeepSeek input was unavailable on all 450 rows. The saved V12 audit made 450 new reasoner calls while replaying two upstream traces. Matched tokens, cost, latency, hardware, and cache telemetry were not retained, so no measured token, dollar, energy, or latency comparison is claimed.

D. Planned Final ExECT Model Comparison

Table IV records the fixed six-model roster. Results remain pending until every condition uses the corrected model-led family architecture.

Historical rows are excluded because Prescription was deterministic-only and SF included an independent extractor union. Corrected candidates remain unpromoted: durable configurations reproduce the attribution and aggregate gates, but deterministic correct-to-wrong counts remain nonzero. DeepSeek uses `deepseek/deepseek-chat` with thinking enabled and is reported as DeepSeek V4 Flash.

TABLE V
SAVED-OUTPUT COMPONENT ABLATION

Component	ExECT dev140	Gan validation750
Normalization/dictionary	+0.0389	+0.0293
Evidence validation	0.0000	0.0000

TABLE VI
RETAINED RELIABILITY EVIDENCE AND LIMITS

Subject	Retained result	Boundary
ExECT evidence	Hybrid dev140 minimum exact evidence rate 1.0000	Development reference.
ExECT calibration	Brier 0.2225; base-rate Brier 0.2340; ECE 0.0587	Full200 aggregate internal scoring-rule result.
ExECT confidence	Test60 failure AUROC 0.5394 GPT; 0.5503 historical DeepSeek; 0.4895 Qwen	Aggregate negative result for saved outputs.
Gan grounding	Architecture-specific validation grounding packages retained	Metrics are not uniform across families.
Reproducibility	1,157 tests, Ruff, mypy, hashes, split restrictions, and six-run replay pass on Python 3.11	Engineering verification, not clinical validation.

E. Cross-Task Components and Reliability

The zero score delta in Table V does not show that evidence validation is unnecessary. A filter may be essential on malformed or hallucinated outputs even when all selected reference predictions already satisfy it. Rejection and repair tests, not aggregate F1 alone, carry that part of the evidence.

Neither predeclared model-confidence review rule met its test60 catch-rate and burden gates, so no confidence-based review policy is adopted. The retained evidence index does not include evidence for a cross-task over-reading claim.

V. DISCUSSION

The reduced system can reproduce six selected runs while keeping component provenance and data limits explicit. The historical ExECT combined control is substantially stronger than the retained rules-only and LLM-only development references on `clinical_headline`, but its Prescription and Seizure Frequency ownership does not meet the final model-led architecture. The three runs also do not share the paper-derived metric views and cannot establish strict benchmark superiority. The rules-only replay shows that CUI matching recovers surface variation, while feature completion remains the larger limitation.

Historical model-transfer artifacts ran with three runtime models, but their Prescription/SF columns and unrecorded DeepSeek thinking state exclude them from the final model comparison. The corrected panel remains incomplete.

The cross-task ablation suggests that normalization is useful in both tasks. The exact-evidence check's zero F1 delta shows why aggregate score changes are insufficient: rejection, source attribution, and repair need direct tests.

VI. LIMITATIONS

Gan test450 contains saved aggregate evidence only and cannot support row-level tuning. ExECT full200 includes dev140 and is not an independent holdout. `clinical_headline` is a clinical fact score, not the published strict benchmark. The paper-derived replay uses development data and does not reproduce the original system, annotation process, or 0.87/0.90 scores. A legacy benchmark/CUI companion has small current-code replay drift. The fixed six-model panel is incomplete, and the historical combined control does not satisfy the final model-led family boundary. Model-reported confidence was uninformative for the three saved historical outputs under the frozen test60 analysis; this is not deployment calibration or a six-model result. Annotation-quality findings are internally adjudicated; unqualified clinical validity would require independent domain review. The current retained evidence does not support a cross-task over-reading claim.

VII. CONCLUSION

An inspectable clinical extraction system can preserve performance evidence and the source needed to interpret it. The selected results support a historical three-method comparison on two tasks and saved Gan holdout results. They expose why the final ExECT model comparison requires a corrected family architecture. They do not support strict ExECT score reproduction, a six-model conclusion, or independent clinical validation. Those boundaries are part of the result, not footnotes to it.

REPRODUCIBILITY STATEMENT

The machine-readable retained evidence index records the six runs, selected files and hashes, policy versions, and no-call replay expectations. The Markdown manuscript and this IEEE source use only selected evidence and claims permitted by the paper claim status.

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